

Roll No.

TEC-601

B. TECH. (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

WIRELESS COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What were the main technical features of the second generation (2G) cellular communication systems ? Discuss some benefits of 2G systems over 1 G system. 10 Marks (CO1)

OR

- (b) Verify that the cluster size is $N = i^2 + j^2 + ij$. 10 Marks (CO1)

2. (a) Assuming 7 cell cluster and UE at the cell boundary, find an expression to indicate SIR in co-channel cells in the downlink. 10 Marks (CO1)

OR

- (b) Find SIR at the boundary of a cell in case of 7, 9, and 12 cells in a cluster. If the minimum required SIR for acceptable voice quality is 18 dB then which one cannot be preferred ? Assume path loss exponent to be 4. 10 Marks (CO1)

P. T. O.

(2)

3. (a) Why sectorization enhances system capacity ? What do you mean by higher order sectorization ? 10 Marks (CO1)

OR

- (b) Discuss different handoff schemes based on the direction.

10 Marks (CO1)

4. (a) Derive an expression to represent the received signal using free space path loss model. 10 Marks (CO2)

OR

- (b) If a transmitter produces 50 watt of power uniformly in all directions, express the transmit power in units of (a) dBm, and (b) dBW. If 50 watt is applied to unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is P_r (10 km) ? Assume unity gain for the receiving antenna. 10 Marks (CO2)

5. (a) State significance of two-ray ground reflection model. Explain using suitable diagrams and obtain an expression for the received signal power. 10 Marks (CO2)

OR

- (b) What do you mean by distorted transmission ? Explain the term constructive and destructive interference. 10 Marks (CO2)

Roll No.

TEC-602

**B. TECH. (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022**

MICROWAVE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Derive field equations for TE_{10} mode in rectangular waveguide.

10 Marks (CO1)

OR

- (b) Guided wavelength of a rectangle waveguide ($2.286 \text{ cm} \times 1.016 \text{ cm}$) is 4.42 cm . When the waveguide is short circuited, find the operating frequency of microwave source (considering it is operating in dominant mode).

10 Marks (CO1)

2. (a) What is excitation of a waveguide ? Explain different methods.

10 Marks (CO1)

OR

- (b) An air filled rectangular waveguide of dimensions 7 cm and 3.5 cm operates in the dominant TE_{10} mode. Find the cutoff frequency and guided wavelength.

10 Marks (CO1)

3. (a) Draw an arrangement to make circulator by using two magic tees.

10 Marks (CO2)

P. T. O.

(2)

OR

(b) Explain the measurement of guided wavelength and frequency, with the help of a block diagram. 10 Marks (CO2)

4. (a) Drive the expressions for S matrix. What is insertion loss and reflection loss? 10 Marks (CO2)

OR

(b) Explain the Hybrid Ring and write the scattering matrix for the four port hybrid ring. 10 Marks (CO2)

5. (a) Explain the working of klystron amplifier with the help of bunching process. 10 Marks (CO3)

OR

(b) A reflex klystron is to be operated at 10 GHz with dc beam voltage 300 V, repeller space 0.1 cm for $1\frac{3}{4}$ mode. Calculate P_{RFmax} and corresponding repeller voltage for a beam current of 20 mA.

10 Marks (CO3)

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TEC-603

B. TECH. (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

VLSI TECHNOLOGY AND DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain silicon shaping. How can we identify the various types of Si wafers. 10 Marks (CO1)

OR

- (b) What do you understand by Epitaxy ? Explain Vapor Phase Epitaxy. Differentiate between Metallurgical Grade Silicon and Electron Grade Silicon. 10 Marks (CO1)

2. (a) What do you understand by Epitaxy ? Explain Vapor Phase Epitaxy. 10 Marks (CO1)

OR

- (b) Explain in brief about CZ Technology and their application.

10 Marks (CO1)

P. T. O.

(2)

3. (a) Explain in brief about ion stopping in VLSI industry.

10 Marks (CO2)

OR

- (b) Explain Mentalization process in VLSI technology. 10 Marks (CO2)

4. (a) With the help of suitable diagram explain various steps of Lithograph.

10 Marks (CO2)

OR

- (b) Explain various types of Dry Etching process in VLSI.

10 Marks (CO2)

5. (a) Explain the Era of VLSI and Y chart. 10 Marks (CO3)

OR

- (b) What is computer added design technology in VLSI ? 10 Marks (CO3)

Roll No.

TEC-604

B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022

DATA COMMUNICATION NETWORK

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain some advantages and disadvantages of combining the session, presentation, and application layer in the OSI model into one single application layer in the Internet model. 10 Marks (CO1)

OR

- (b) Write short notes on the following : 10 Marks (CO1)

(i) Wide Area Network (WAN)

(ii) Mesh Topology

2. (a) Calculate the CRC for the data word 1010101010 and divisor be 1001 which is agreed upon by the sender and the receiver. Show the steps performed at sender and destination end. 10 Marks (CO2)

OR

- (b) Explain the basic function of data link layer in the OSI model.

10 Marks (CO2)

P. T. O.

(2)

3. (a) How do the layers of the TCP/IP Internet model correlate to the layers of the OSI model ? Explain. 10 Marks (CO1)

OR

- (b) Describe the goals of multiplexing. Explain Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM) in brief.

10 Marks (CO1)

4. (a) Explain the various services provided by the data link layer to the network layer. 10 Marks (CO2)

OR

- (b) Describe sliding-window flow control. What are the advantages of sliding-window flow control compared to stop-and-wait flow control.

10 Marks (CO2)

5. (a) List and explain in all the line coding schemes used for digital transmission of signals over a data communication network.

10 Marks (CO1, CO2)

OR

- (b) What do you understand by HDLC protocol ? 10 Marks (CO1, CO2)

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TOE-610

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

OOPs USING C++

Time : 1½ Hours

Maximum Marks : 50

- Note:** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What is the significance of private, protected, and public specifiers in a class ? Explain with the help of example. 10 Marks (CO1, CO2)

OR

- (b) Write a C++ program to create a class called TaskDemo with the following members : 10 Marks (CO1, CO2)

Data Members :

string str

Function Members :

Constructor to initialize value of str

void printvalue () → This member function will print value of str

void printchars () → This member function will print all even position characters from the data member str

In main function create object of class TaskDemo and perform the above task.

P. T. O.

2. (a) Implement with a C++ program to input 3×5 matrix and count and print number of palindrome numbers row-wise. 10 Marks (CO1, CO2)

OR

- (b) Illustrate different types of Constructors in C++ with proper code. 10 Marks (CO1, CO2)

3. (a) Differentiate between of the following : 10 Marks (CO1, CO2)
- (i) Call by value and Call by reference
 - (ii) Break and continue
 - (iii) Static data member and instance data member

OR

- (b) Describe static binding in C++ with an example. 10 Marks (CO1, CO2)
4. (a) What is object-oriented programming 4 characteristics ? Explain advantages of object-oriented programming paradigm over structured programming paradigm. 10 Marks (CO1, CO2)

OR

- (b) Define a class named MidDemo with following description : 10 Marks (CO1, CO2)

Data Members

private int num

Constructor MidDemo (int) → Constructor will store value in num.

Member function printval () → this member function will print value of num.

Friend function void reverse (MidDemo ob) → This friend function will reverse the number.

Example

If value of num is 7456

Then after reverse value of num is 6547.

(3)

5. (a) What is the difference between static and non-static member function in C++ ? Explain with the help of C++ program. 10 Marks (CO1, CO3)

OR

- (b) With the help of C++ program explain any *three* uses of scope resolution (::) operator. 10 Marks (CO1, CO3)

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TEC-659

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

ADVANCED EMBEDDED SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Differentiate between general purpose system and embedded systems.

10 Marks (CO1)

OR

- (b) Explain the classification of Embedded System and discuss about the applications of it.

10 Marks (CO1)

2. (a) Explain the quality attributes of Embedded System.

10 Marks (CO1)

OR

- (b) How can we design a currency counter using any microcontroller ?
Explain in detail about the designing process of it.

10 Marks (CO1)

P. T. O.

(2)

3. (a) Explain PIC microcontroller with the help of a block diagram.

10 Marks (CO2)

OR

- (b) Explain 8051 internal architecture and features of 8051 microcontrollers.

10 Marks (CO2)

4. (a) How are PC and SP affected while using CALL instructions ? Explain the difference between ACALL and LCALL instructions.

10 Marks (CO2)

OR

- (b) WAP to add two 16 bit numbers FF26H and 2289 H using 8051. What effect will be on PSW after addition ?

10 Marks (CO3)

5. (a) Explain the timers of 8051. What is the role of Gate bit ? Discuss the details of registers used for Timer of 8051.

10 Marks (CO3)

OR

- (b) A LED is connected to P0.1. Using timer create delay of 250 ms and blink the led in this duration.

10 Marks (CO3)